

Question	Marking Guidance	Mark	Comments
01.1	1. Substrate binds to the active site/enzyme OR Enzyme-substrate complex forms; 2. Active site changes shape (slightly) so it is <u>complementary</u> to substrate OR Active site changes shape (slightly) so distorting/breaking/forming bonds in the substrate; 3. Reduces activation energy;	3	1. Accept for 'binds', fits

Question	Marking Guidance	Mark	Comments
01.2	Adenosine diphosphate;	1	

Question	Marking Guidance	Mark	Comments
01.3	Mark in pairs, 1 and 2 OR 3 and 4 OR 5 and 6 1. Boil OR Add (strong) acid/alkali; 2. Denatures the enzyme/ATP synthase; OR 3. Put in ice/fridge/freezer; 4. Lower kinetic energy so no enzyme-substrate complexes form; OR 5. Add high concentration of inhibitor; 6. Enzyme-substrate complexes do not form;	2	1. Accept heat at > 50°C OR at very high temperatures 2. Accept for 'denatures', a description of denaturation 4. Accept ES for enzyme substrate complex

Question	Marking Guidance	Mark	Comments
01.4	1. (With) increasing Pi concentration, more enzyme-substrate complexes are formed; 2. At or above 40 (mmol dm⁻³) all active sites occupied OR At or above 40 (mmol dm⁻³) enzyme concentration is a limiting factor;	2	

